15: Performance Profiling

Engr 315: Hardware / Software Codesign

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Some material taken from EECS370 at U. of Michigan

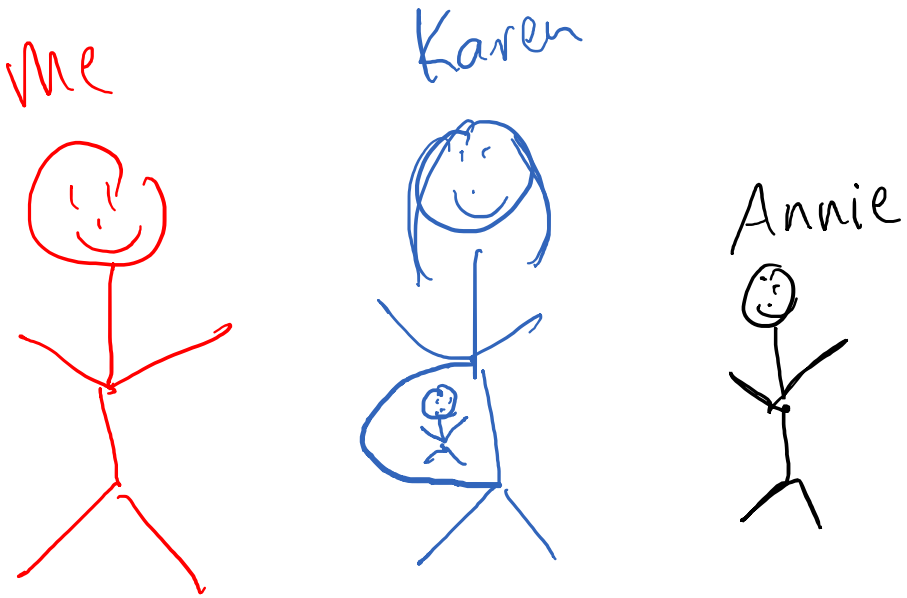
The COVID-19 Plan

- Adding remote lecture streaming

- Current Plan:
- I'll still be in 4111 on M/W @ 4.

- Plans may evolve

Also in April…



P4

- Count the number of binary 1's in a file.

- Goal: <5 seconds
- My best: 0.8049 seconds
- Ethan: 0.4?? seconds

Where to next?

- Less “How does it work”
- More “Make it work *faster*”

Last Time

```
# 'medium.bin' contains 1M of random 1's
print("Timing 'medium.bin'")
time = tt.timeit(count_medium_sw, number=1)
print("Total Time:" + str(time) + " seconds")
```

```
Timing 'medium.bin'
Found 4196661 Ones
Total Time:51.849690064002 06 seconds
```

What’s taking 51 seconds?

- My guess:

```
def countInt(self, n):
    w = 0
    while (n):
        w += 1
        n &= n - 1
    return w
```

- How can I **know**?

Code Profiling

- In software engineering, profiling ("program profiling", "software profiling") is a form of dynamic program analysis that measures, for example, the space (memory) or time complexity of a program, the usage of particular instructions, or the frequency and duration of function calls. Most commonly, profiling information serves to aid program optimization. [Wiki]

Code Profiling can measure

- Program Runtimes
- Function Call Numbers/Runtimes
- Memory Usage
- Instruction Usage
- Others

Common Profilers

- Function Call Numbers/Runtimes
  - C/C++: gprof
  - Python: cProfile

- Memory Usage
  - Valgrind

Python's cProfile

- cProfile and profile provide *deterministic profiling* of Python programs. A *profile* is a set of statistics that describes how often and for how long various parts of the program executed.

- https://docs.python.org/3/library/profile.html

Python’s cProfile

```python
import cProfile
import re
cProfile.run('re.compile("foo|bar")')
```

```
      5 function calls in 0.000 seconds

Ordered by: standard name

ncalls  tottime  percall  cumtime  percall filename:lineno(function)
     1    0.000    0.000    0.000    0.000 <string>:1(<module>)
     1    0.000    0.000    0.000    0.000 re.py:231(compile)
     1    0.000    0.000    0.000    0.000 re.py:286(_compile)
     1    0.000    0.000    0.000    0.000 {built-in method builtins.exec}
     1    0.000    0.000    0.000    0.000 {method 'disable' of '_lsprof.Profiler' objects}
```

cProfile's Output Explained

Number of calls

Total Time spend in the given function (excluding sub-functions)

`tottime/ncalls`

Cumulative time spend in this and all sub-functions

`cumtime/ncalls`

Function name

```
ncalls  tottime  percall  cumtime  percall filename:lineno(function)
     1    0.000    0.000    0.000    0.000 <string>:1(<module>)
```

cProfile on Software_Ones_Counter

```
cProfile.run('count_medium_sw()')
```

```
         262189 function calls in 54.148 seconds

   Ordered by: standard name

   ncalls  tottime  percall  cumtime  percall filename:lineno(function)
        1    0.000    0.000   54.146   54.146 <ipython-input-5-9fda7f5f3069>:11(countFile)
   262144   51.267    0.000   51.267    0.000 <ipython-input-5-9fda7f5f3069>:21(countInt)
        1    2.820    2.820   54.087   54.087 <ipython-input-5-9fda7f5f3069>:28(countArray)
        1    0.001    0.001   54.148   54.148 <ipython-input-7-a468c2b9dd06>:11(count_medium_sw)
        1    0.000    0.000   54.148   54.148 <string>:1(<module>)
        1    0.000    0.000    0.000    0.000 _bootlocale.py:23(getpreferredencoding)
        1    0.000    0.000    0.000    0.000 codecs.py:259(__init__)
        1    0.000    0.000    0.000    0.000 codecs.py:308(__init__)
        1    0.000    0.000    0.000    0.000 codecs.py:334(setstate)
```

cProfile on Software_Ones_Counter

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cProfile.run('count_medium_sw()')
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   Ordered by: standard name

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        1    0.000    0.000   54.148   54.148 <string>:1(<module>)
        1    0.000    0.000    0.000    0.000 _bootlocale.py:23(getpreferredencoding)
        1    0.000    0.000    0.000    0.000 codecs.py:259(__init__)
        1    0.000    0.000    0.000    0.000 codecs.py:308(__init__)
        1    0.000    0.000    0.000    0.000 codecs.py:334(setstate)
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cProfile on Software_Ones_Counter

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cProfile.run('count_medium_sw()')
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        1    0.000    0.000   54.148   54.148 <string>:1(<module>)
        1    0.000    0.000    0.000    0.000 _bootlocale.py:23(getpreferredencoding)
        1    0.000    0.000    0.000    0.000 codecs.py:259(__init__)
        1    0.000    0.000    0.000    0.000 codecs.py:308(__init__)
        1    0.000    0.000    0.000    0.000 codecs.py:334(setstate)
```

cProfile on Software_Ones_Counter

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cProfile.run('count_medium_sw()')
```

```
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   Ordered by: standard name

   ncalls  tottime  percall  cumtime  percall filename:lineno(function)
        1    0.000    0.000   54.146   54.146 <ipython-input-5-9fda7f5f3069>:11(countFile)
   262144   51.267    0.000   51.267    0.000 <ipython-input-5-9fda7f5f3069>:21(countInt)
        1    2.820    2.820   54.087   54.087 <ipython-input-5-9fda7f5f3069>:28(countArray)
        1    0.001    0.001   54.148   54.148 <ipython-input-7-a468c2b9dd06>:11(count_medium_sw)
        1    0.000    0.000   54.148   54.148 <string>:1(<module>)
        1    0.000    0.000    0.000    0.000 _bootlocale.py:23(getpreferredencoding)
        1    0.000    0.000    0.000    0.000 codecs.py:259(__init__)
        1    0.000    0.000    0.000    0.000 codecs.py:308(__init__)
        1    0.000    0.000    0.000    0.000 codecs.py:334(setstate)
```

cProfile on Software_Ones_Counter

```
cProfile.run('count_medium_sw()')
```

```
      262189 function calls in 54.148 seconds

 Ordered by: standard name

 ncalls  tottime  percall  cumtime  percall filename:lineno(function)
      1    0.000    0.000   54.146   54.146 <ipython-input-5-9fda7f5f3069>:11(countFile)
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      1    0.000    0.000    0.000    0.000 codecs.py:259(__init__)
      1    0.000    0.000    0.000    0.000 codecs.py:308(__init__)
      1    0.000    0.000    0.000    0.000 codecs.py:334(setstate)
```
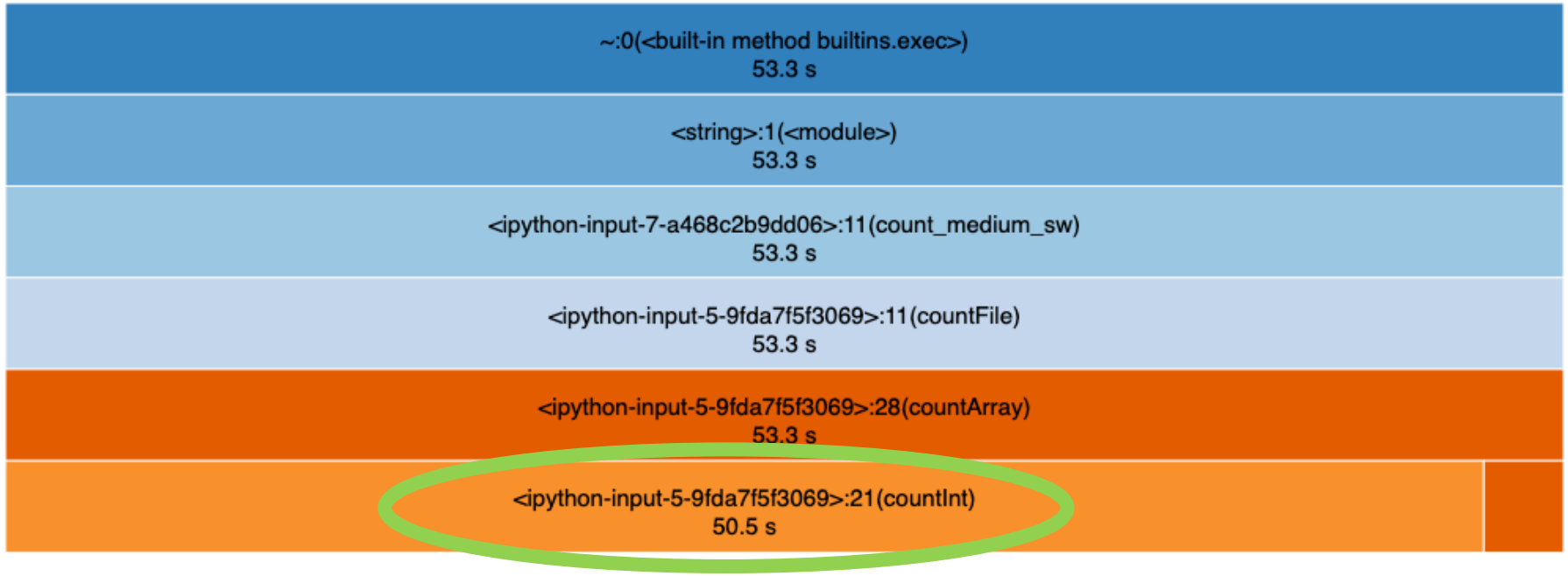

cProfile + SnakeViz

```
cProfile.run('count_medium_sw()', 'sw.cprof')
```

```
$ snakeviz sw.cprof
```



```python
cProfile.run('count_large_hw()')
```

```
ncalls  tottime  percall  cumtime  percall filename:lineno(function)
     1    0.000    0.000    0.883    0.883 <ipython-input-5-9fda7f5f3069>:11(countFile)
     1    0.038    0.038    0.836    0.836 <ipython-input-6-c3ed96e08cf2>:28(countArray)

   641    0.055    0.000    0.249    0.000 dma.py:104(transfer)
     1    0.000    0.000    0.000    0.000 dma.py:130(wait)
   642    0.010    0.000    0.033    0.000 dma.py:65(running)
   642    0.008    0.000    0.025    0.000 dma.py:72(idle)

  1285    0.034    0.000    0.039    0.000 mmio.py:114(read)
  1283    0.047    0.000    0.051    0.000 mmio.py:144(write)
  2568    0.009    0.000    0.009    0.000 mmio.py:182(_debug)
```

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